

Complete Technical Architecture & Technology Stack Specification

Smart India Hackathon 2026 • Problem Statement: SIH26056 • Ministry of Statistics (MoSPI) & DGCA

Executive Architecture Summary: AeroDex is a cloud-ready, dual-layer econometric platform built in Python. It pairs ethical multi-source web extraction (Playwright, HTTP SSR, Robots.txt enforcement) with DGCA Form-A census weights (786 domestic routes, 136M annual flyers) to compute real-time Laspeyres, Paasche, and Fisher superlative airfare indices.

1. CORE LANGUAGE, RUNTIME & SERVER INFRASTRUCTURE

Backend Engine & Runtime

- Python 3.12+CPythonAsyncio
- Asynchronous Concurrency:** Native `asyncio` event loop with `asyncio.gather()` for multi-source concurrent OTA scraping.
 - Thread-Safe Operations:** Re-entrant thread locks (`threading.Lock`) protecting in-memory caches and SQLite connection pools.

Web & REST API Gateway

- http.serversocketserverJSON REST
- Zero External Framework Lock-in:** Lightweight native HTTP server operating with high resilience and minimal overhead.
 - CORS & Security Headers:** Full CORS negotiation, mime-type streaming, CSV exports, and dynamic port binding (PORT env variable).

2. WEB AUTOMATION & ETHICAL SCRAPING STACK

Headless Browser Automation

- Playwright 1.40+Chromiumplaywright-stealth
- Stealth Evasion:** Overrides `navigator.webdriver` and Chromium fingerprints to evade bot challenges.
 - Resource Interception:** Aborts media/font/image requests (`.png` , `.woff` , `.mp4`) saving >70% bandwidth with RAM <120MB.
 - Target Portals:** Google Flights, EaseMyTrip, Cleartrip, and SpiceJet Direct Portal.

Ethical Compliance & Governance

- urllib.robotparserRFC 9309RobotGuard
- Pre-Request Robots.txt Check:** Evaluates domain permissions prior to dispatching network requests.
 - Polite Crawl Rate Limiter:** Enforces 3.0s per-domain crawl delays and exponential backoff on HTTP 429/503.
 - Ethical Gating:** Aborts scraping on Disallow rules (Ixigo) and generates 1-click verification links instead.

3. ECONOMETRIC, MATHEMATICAL & CPI INDEX ENGINE

Index Formulations & Weights

- Laspeyres (IL)Paasche (IP)Fisher (IF)
- DGCA Spatial Route Weights (\$W_r\$):** Form-A passenger volume weights across 786 domestic routes (136M flyers).
 - Carrier Share Aggregation (\$\omega_c\$):** IndiGo (~62%), Air India (~14.5%), Akasa (~5.5%), SpiceJet (~4.5%), AIX (~8%).
 - MoSPI Headline CPI Linkage:** Quantifies exact basis point impact on CPI Sub-class 07.3.3 (0.077% basket weight).

Statistical Data Cleaning & Audit

- Tukey 1.5x IQRSHA-256Tariff Deconstruction
- Fare Decomposition:** Deconstructs Base (~70%), Fuel YQ (~18%), Airport UDF/PSF (~7%), and GST (5%).
 - Outlier Filtering:** 2.5× median IQR cap eliminates luxury skew.
 - Data Integrity Engine:** Computes 4D composite quality score (0-100%, Grade A+) sealed with SHA-256 cryptographic hashes.

4. MACHINE LEARNING, NOWCASTING & REAL-TIME SENSOR TELEMETRY

ML Nowcaster & Vectorized Models

scikit-learn 1.3+ joblib NumPy

- **RandomForestRegressor:** Pre-trained model (`airfare_predictor.joblib`) predicting booking horizon surges (\$T+0\$ to \$T+30\$).
- **Feature Pipeline:** Vectorized features including distance, days ahead, load factor estimates, holiday flags, and fuel index.

Live Calamity & Weather Telemetry

NOAA / NWS METAR UN GDACS API BeautifulSoup4

- **Airport Weather Sensors:** Real-time METAR parser extracting visibility, wind, and fog conditions for Indian hubs.
- **Disaster Tracker:** Ingests UN Global Disaster Alert RSS/JSON feeds to model emergency airfare surge dynamics.

5. AUTONOMOUS AI POLICY & SITUATION ROOM

Google Gemini GenAI Engine

google-genai Gemini 3.6 Flash JSON Schema

- **Executive Briefings:** Generates automated macroeconomic summaries for RBI MPC and MoSPI officials.
- **Bilingual Generation:** Synthesizes executive briefings in English and formal Hindi (शुद्ध हिन्दी).
- **Deterministic Fallback:** Local template engine guarantees 100% uptime when API keys or network are offline.

Historical Shock Replay Studio

Stress-Testing Chained Substitution

HHI Antitrust

- **Historical Aviation Crises:** Replays Go First Grounding (2023), Jet Airways Collapse (2019), and ATF Fuel Spike (2022).
- **Anti-Trust Watchdog:** Computes Herfindahl-Hirschman Index (HHI) for carrier route monopoly surveillance (DGCA Rule 135 & CCI).

6. STORAGE WAREHOUSE & FRONTEND DASHBOARD STACK

Microdata Warehouse Tier

SQLite 3 WAL Mode CSV Data Feeds

- **Audit Warehouse:** Microdata table recording granular flight observations, timestamps, carrier codes, and provenance tags.
- **MoSPI Bulletins:** Dynamic export endpoints for Daily Sector Bulletins, Weekly Aggregation, and Microdata Archives.

Interactive UI & Visualization

HTML5 / Vanilla CSS3 ES6+ JavaScript Chart.js

- **Design Aesthetics:** Dark glassmorphism executive theme, responsive grid, dynamic pulse status telemetry.
- **Real-time Visualizations:** High-frequency index line charts, HHI gauge meters, route weight distributions.

7. TECHNOLOGY STACK MATRIX & LIBRARY BREAKDOWN

Category	Technology / Library	Version	Primary Role in AeroDex Architecture
Core Runtime	Python (CPython)	≥ 3.12	Central execution runtime, thread management, async event loop
Web Server	Python Standard Library (http.server)	Built-in	REST API routing, microdata JSON serialization, static asset delivery
Scraping Engine	Playwright	≥ 1.40.0	Headless Chromium browser automation across EaseMyTrip, Cleartrip & SpiceJet
Anti-Bot Defense	playwright-stealth	≥ 2.0.0	Evasion of WebGL/navigator automation fingerprints
Compliance	urllib.robotparser (RobotGuard)	Built-in / RFC 9309	Pre-request robots.txt compliance validation and rate limit governance
HTTP Networking	Requests & BeautifulSoup4	≥ 2.31 / ≥ 4.12	Ultra-fast HTTP SSR scraping of Google Flights & NOAA METAR telemetry
Machine Learning	scikit-learn & joblib	≥ 1.3.0	Pre-trained Random Forest nowcasting model and inference pipeline
Numerical Ops	NumPy	≥ 1.24.0	Vectorized econometric calculations, outlier IQR filtering, HHI metrics
Generative AI	Google Gemini (google-genai)	≥ 0.1.0	Gemini 3.6 Flash Autonomous AI Situation Room policy briefings
Database	SQLite3 (with WAL mode)	Built-in	Persistent microdata quote audit warehouse and time-series quotes
Frontend UI	HTML5, Vanilla CSS3, ES6+ JS	Native Standards	High-performance executive dashboard with glassmorphism styling
Charts & Graphics	Chart.js	≥ 4.4.0 (CDN)	Real-time price trend charts, shock comparison graphs, and HHI gauges
Typography	Google Fonts (Inter, JetBrains Mono)	Web Fonts	Executive typography with high legibility for data-dense tables
DevOps / PaaS	Render, Railway, Docker, Procfile	Standard	Multi-cloud containerization, automatic PORT binding, and health checks
Testing	unittest (Python standard library)	Built-in	Automated test suite (31 tests covering math, weighting & APIs)